

▷ every refusal releases the claims taken so far

1: **procedure** PROCESSREQUEST(r)
2: $p \leftarrow \text{ATTEST}(r)$; **if** $r.\text{address}$ is a proxy **or** p invalid **then refuse**
▷ Step 0: App Attest or Play Integrity; the platform header is ignored
3: **if** session(r) unsigned, not bound to $r.\text{fingerprint}$, nonce replayed, **or** 3 requests in 10 min **then refuse**
▷ Step 1
4: **if** $r.\text{fingerprint}$ **or** $r.\text{address}$ is denylisted **then refuse**
5: **for** $k \in \{\text{address}, /24, \text{ASN}\}$ **do if not** ATOMICTAKE(limit $_k$) **then refuse**
▷ Step 2: one Redis operation per limit; lower caps for hosting ASNs
6: **if** $p = \text{web}$ **and** (reCAPTCHA score $< \theta_r$ **or** origin $\notin$ allowed) **then refuse**
▷ Steps 3–4
7: $m \leftarrow \text{NORMALIZE}(r.\text{number})$; **if** country(m) $\notin$ allow-list **then refuse**
8: signals $+=$ cost class of prefix(m) (premium: **refuse**), sequential or narrow-range burst, VoIP; **if** HLR says dead **then refuse**
9: $v \leftarrow \text{BLOCKVERDICT}(\text{block}(m))$ ▷ the stage in force on the 8-digit block
▷ Step 5: the bulk flag needs an internal credential
10: **if** body invalid **or** m excluded **then refuse** ▷ Step 6: the test flag needs a credential
11: $s \leftarrow \text{Equation}(1)$; $t \leftarrow \text{TIER}(s, \text{mode})$ ▷ Step 7: allow, delay, challenge, downgrade, block
12: **if** r has no verified history: $t \leftarrow \max(t, \text{challenge})$ **if** $v = 1$, $\max(t, \text{downgrade})$ **if** $v = 2$
13: **if** $t = \text{block}$ **then refuse**; **if** $t = \text{challenge}$ **and** $p = \text{web}$ **then return** challenge
▷ the client retries with the solved challenge, which scores -20
14: **if not** ATOMICCLAIM(m : window 60 s · 2^{sends} up to 1 h; 5 a day) **then refuse** ▷ Step 8
15: $\mu \leftarrow \text{ADAPTIVEMULTIPLIER}(\text{source}, p, \text{country}) \in [0.25, 1.5]$ ▷ Step 9: from the baseline job
16: **if not** ATOMICTAKE(source caps $\times \mu$) **then refuse** ▷ verified history keeps the static cap
17: mode $\leftarrow \text{OPERATINGMODE}(\text{hourly count}, \text{spend})$ ▷ Step 10: elevated at 80%, emergency at 100%
18: **if not** ATOMICRESERVE(hourly count, spend, block(m)’s counter slot) **then refuse** ▷ Step 11
19: **if** the slot is past the quota **and** r has no verified history: $t \leftarrow \max(t, \text{challenge})$; past twice: $\max(t, \text{downgrade})$
20: channel $\leftarrow$ SMS, or push or WhatsApp **if** $t = \text{downgrade}$; write state; enqueue(m , channel, delay **if** $t = \text{delay}$)
21: **return** one uniform HTTP 200 body after at least 400 ms ▷ the same body for sent, dropped and refused

22: **procedure** ONOUTCOME(send, outcome) ▷ asynchronous; outcome $\in \{\text{verified}, \text{failed}, \text{undelivered}\}$
23: record the transition and its effects in one compare-and-set; apply each effect at most once
▷ failed: five wrong codes or 2 min after delivery; undelivered: no receipt in 60 s, or a failed receipt
24: **for** $k \in \text{keys}(\text{send})$ **do** ▷ address, /24, ASN, fingerprint, session, country, prefix, number
25: verified $_k$ or failed $_k$ $+= 1$ ▷ undelivered counts toward neither
26: **if** k is a fine key, resolved $_k \geq 50$ **and** verified $_k$ /resolved $_k < 0.1$ **then** denylist k for 24 h
27: **if** verified **then** mark the number trusted ▷ verified history exempts the client from both destination policies
28: $b \leftarrow \text{block}(\text{number})$; conv $_b \leftarrow \max(-ch, \text{conv}_b + (\text{verified?} - 2.08 : +1.50))$
29: **if** verified **then** speed $_b \leftarrow \max(-ch, \text{speed}_b + (\text{entered within 5 s?} + 1.50 : -2.08))$
30: **if** conv $_b > h$ (after 3 events) **or** speed $_b > h$ **then** issue a verdict for 1 h and reset both statistics
▷ stage 1, or stage 2 if a verdict is active: Section 3.5; $h = \ln 1000$, c the floor parameter